REPRESENTATION LEARNING FOR NUCLEIC ACID SEQUENCES

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TODAY'S PRACTICAL

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WHAT WILL YOU LEARN?

Embed sequences to numeric representations and extract useful features.

Combine and manipulate these features to create feature spaces.

Implement various models and evaluations to identify important features.

Apply the above on different use cases.

DATASETS: DNABERT2

Species	Task	Num. Datasets	Num. Classes	Sequence Length
Human	Core Promoter Detection	3	2	70
	Transcription Factor Prediction	5	2	100
	Promoter Detection	3	2	300
	Splice Site Detection	1	3	400
Mouse	Transcription Factor Prediction	5	2	100
Yeast	Epigenetic Marks Prediction	10	2	500
Virus	Covid Variant Classification	1	9	1000

WORKFLOW: SEQUENCE CLASSIFICATION

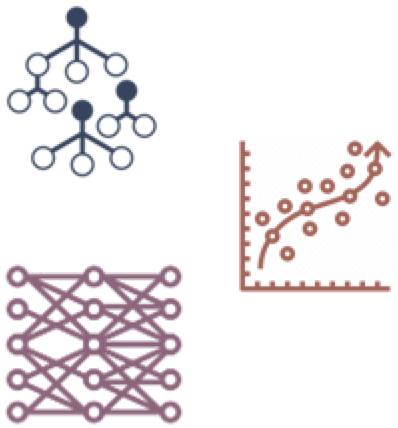
28 Datasets
4 Species
7 Use cases

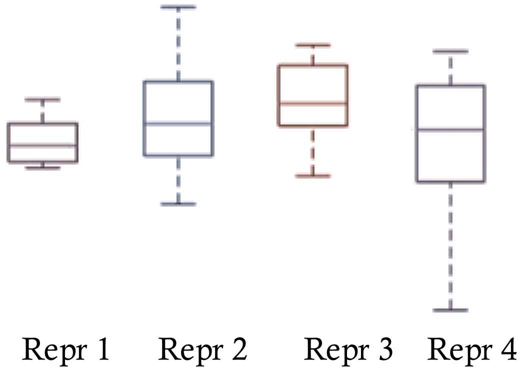
Sequence	Label

7 Representations

Repr 1	Repr 2	Repr 3	Repr 4

5 Models





Code: [GitHub Link]

SEQUENCE REPRESENTATION

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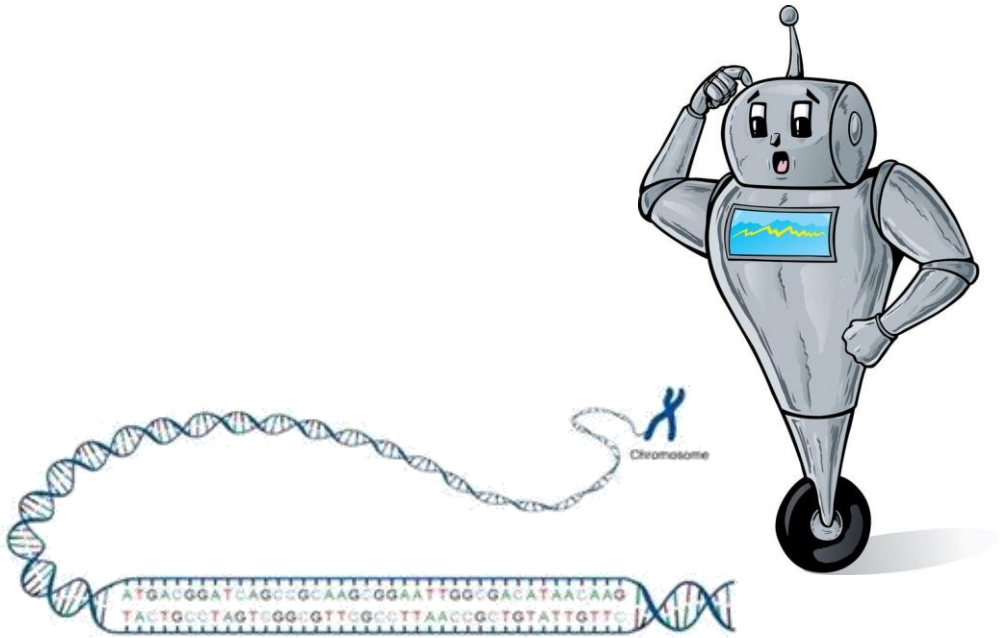
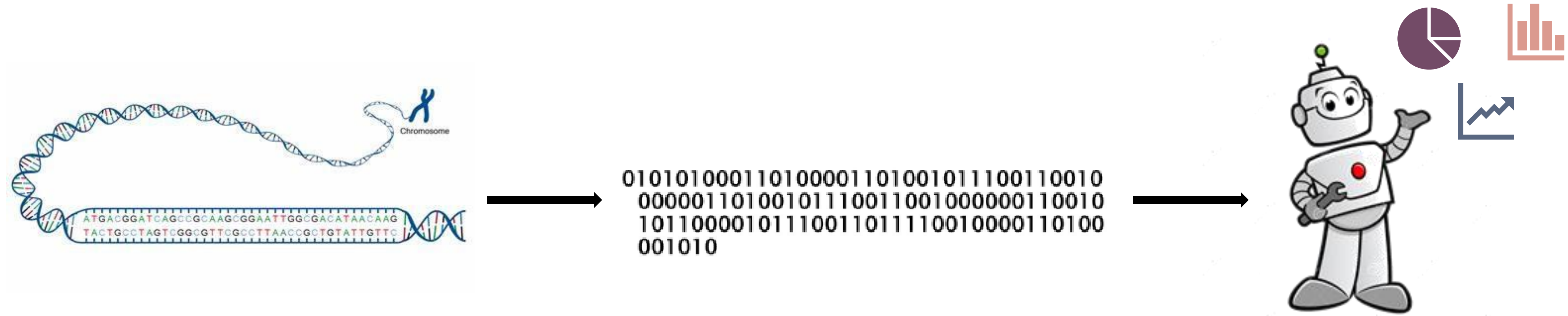
WHY DO WE NEED SEQUENCE REPRESENTATIONS?



Computers don’t read letters!

WHAT IS REPRESENTATION LEARNING?



•**It transforms raw sequences into useful numeric forms** (vectors or embeddings) that capture patterns, structure, or biological meaning.

•**Helps in tasks like gene prediction, classification, or motif detection** without manually crafting features like k-mers or GC content.

TYPES OF REPRESENTATIONS

Basic Encodings

Physico-chemical Features

Numerical Mapping

Complex Representations

BASIC ENCODINGS

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ONE-HOT ENCODINGS

sequence

	G	G	A	T	A	C
A						
T						
G						
C						

What

Binary vector representing each nucleotide

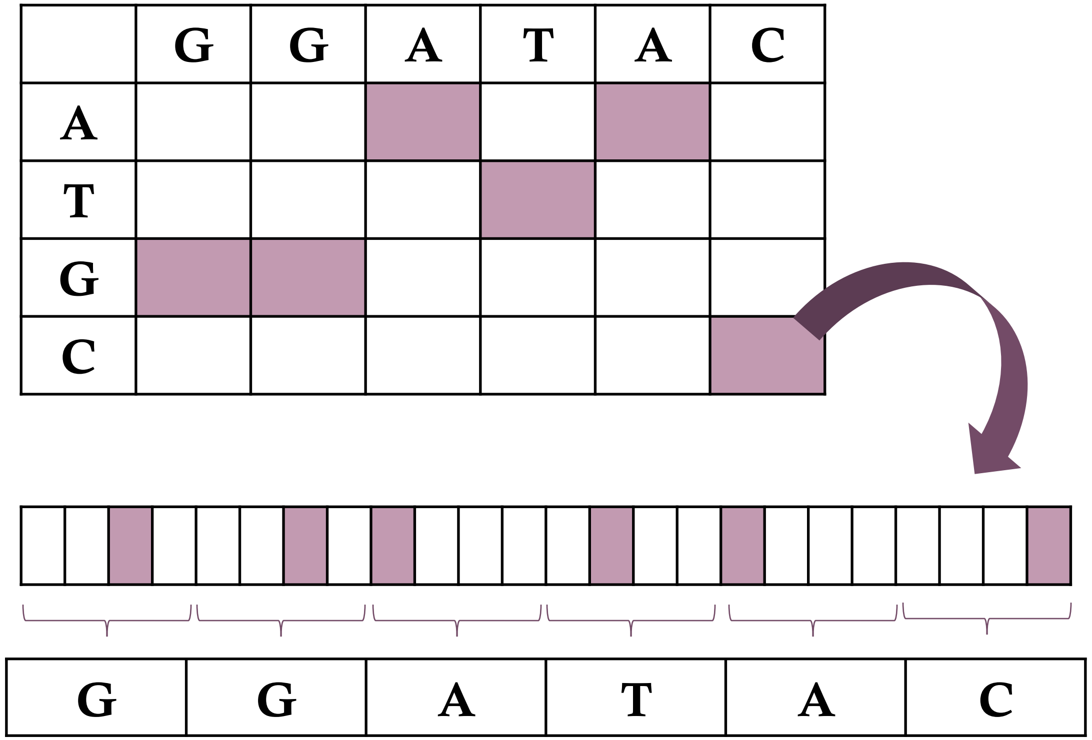
When

Deep learning, where fixed input formats are needed.

Drawbacks

High dimensionality, ignores sequence context, no similarity captured.

ONE-HOT ENCODINGS



What

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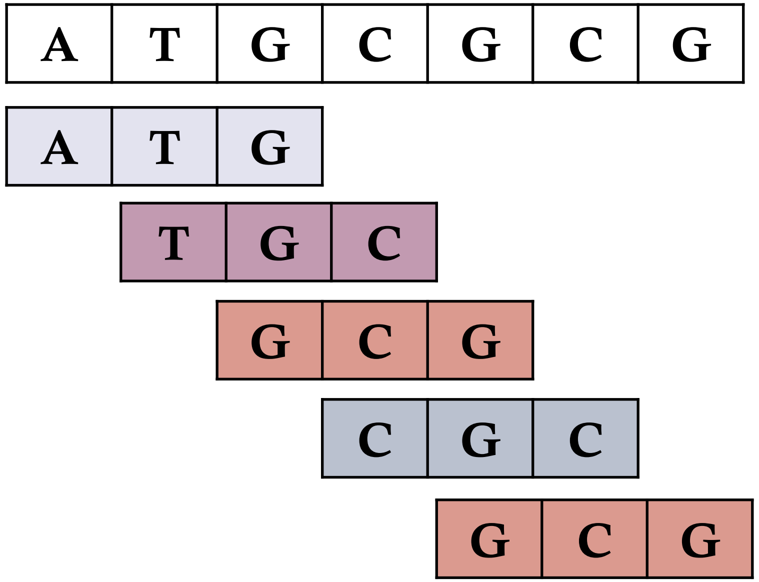
K-MER



K-mer	Count
ATG	1
TGC	1
GCG	2
CGC	1

What

Counts frequency of sub-sequences of length k

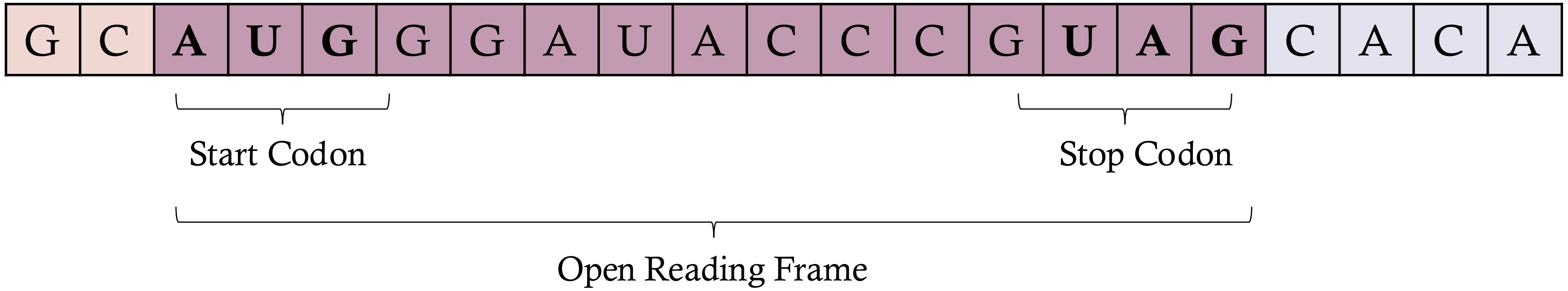
When

Feature engineering for ML models, sequence comparison.

Drawbacks

Loses global order, large k leads to sparse high-dimensional space.

OPEN READING FRAMES



What

Encodes information about potential protein-coding regions (length, position).

When

Gene prediction, protein-coding region identification.

Drawbacks

Only applicable to coding sequences; may miss short/atypical ORFs.

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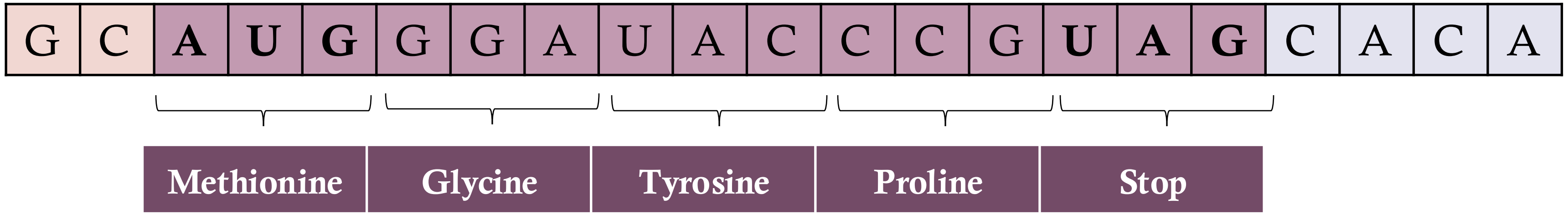

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PHYSICO-CHEMICAL DESCRIPTORS

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GC CONTENT

G	C	A	T	G	G	G	A	T	G	C	C	G	T	A	G	C	A	C	A

60%

What

Proportion of G and C nucleotides.

When

Genome classification, stability prediction.

Drawbacks

Very coarse; ignores sequence order and context.

NUCLEOTIDE RELATED

G	C	A	T	G	G

🧬 **Dinucleotide Features**

- Dinucleotide frequency
- Transition/Transversion ratio
- Position-specific occurrence

🧬 **Trinucleotide Features**

- Trinucleotide frequency
- Codon usage bias
- GC content in trinucleotides

What

Composition (dinucleotide/trinucleotides), physicochemical properties of nucleotides.

When

DNA/RNA function prediction, motif analysis.

Drawbacks

May be too simplified or too detailed depending on feature choice.

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GLOBAL DESCRIPTORS

	G	C	A	T	G	G	
A			1				0.17
T				1			0.17
G	1				2	3	0.5
C		1					0.17

What

Cumulative nucleotide composition.

When

Classification tasks.

Drawbacks

May lose local sequence information.

NUMERIC MAPPINGS

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REAL

G	C	A	T	G	G

A	**0.5**
T	**-0.5**
G	**1.5**
C	**-1.5**

1.5	-1.5	0.5	-0.5	1.5	1.5

What

Assigns real numbers to nucleotides.

When

Mathematical modeling, signal processing.

Drawbacks

Arbitrary mappings can affect results, ignores biological semantics.

INTEGERS

G	C	A	T	G	G

A	0
T	1
G	2
C	3

2	3	0	1	2	2

What

Encodes nucleotides as integers.

When

Simple ML models, sequence alignment, indexing.

Drawbacks

Imposes false ordinal relationships; lacks biological meaning.

Z-CURVE

	X (Purine vs Pyrimidine)	Y (Amino vs Keto)	Z (Strong vs Weak Bond)
A	1	1	1
T	-1	-1	1
G	1	-1	-1
C	-1	1	-1

G	C	A	T	G	G

1	-1	-1	-1	1	-1	1	1	1	-1	-1	1	1	-1	-1	1	-1	-1
1	**-1**	**-1**	**0**	**0**	**-2**	**1**	**1**	**-1**	**0**	**0**	**0**	**1**	**-1**	**-1**	**2**	**-2**	**-2**

What

3D curve representation based on nucleotide distribution.

When

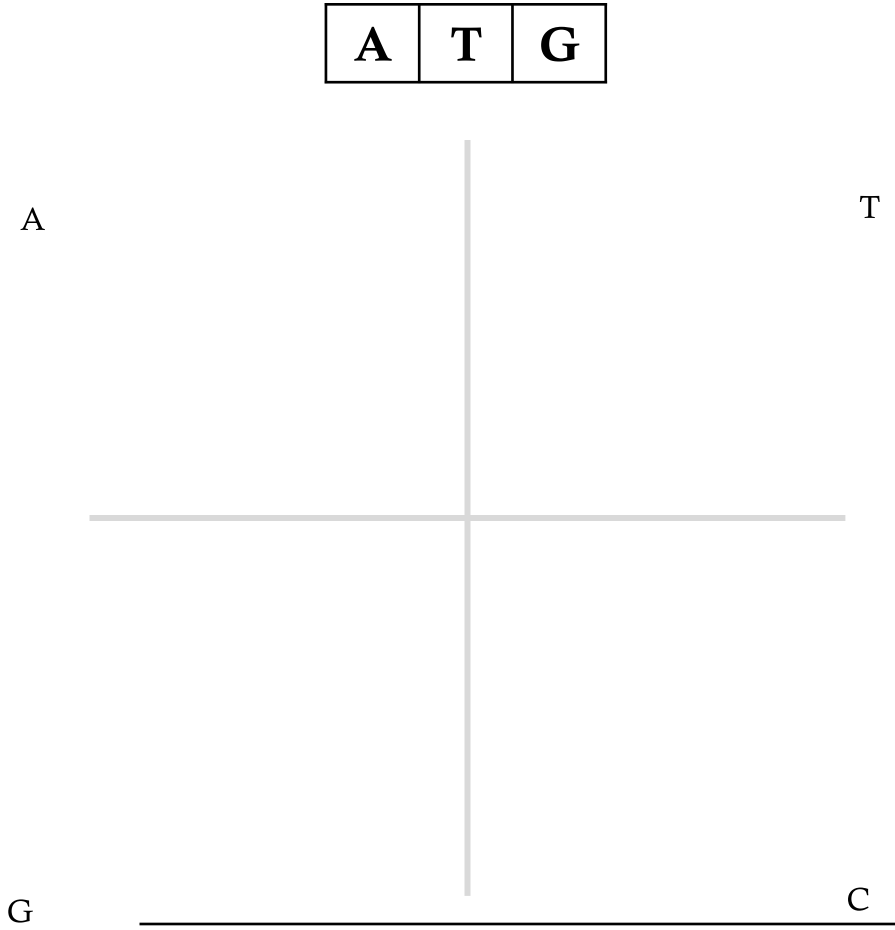
Feature engineering for ML models, genome comparison, gene identification..

Drawbacks

Complex to compute and interpret.

MORE COMPLEX EMBEDDINGS

CHAOS GAME



What

Fractal-like image encoding sequence as 2D patterns.

When

Visual pattern recognition, deep learning on images.

Drawbacks

Computationally heavy for long sequences; hard to interpret directly.

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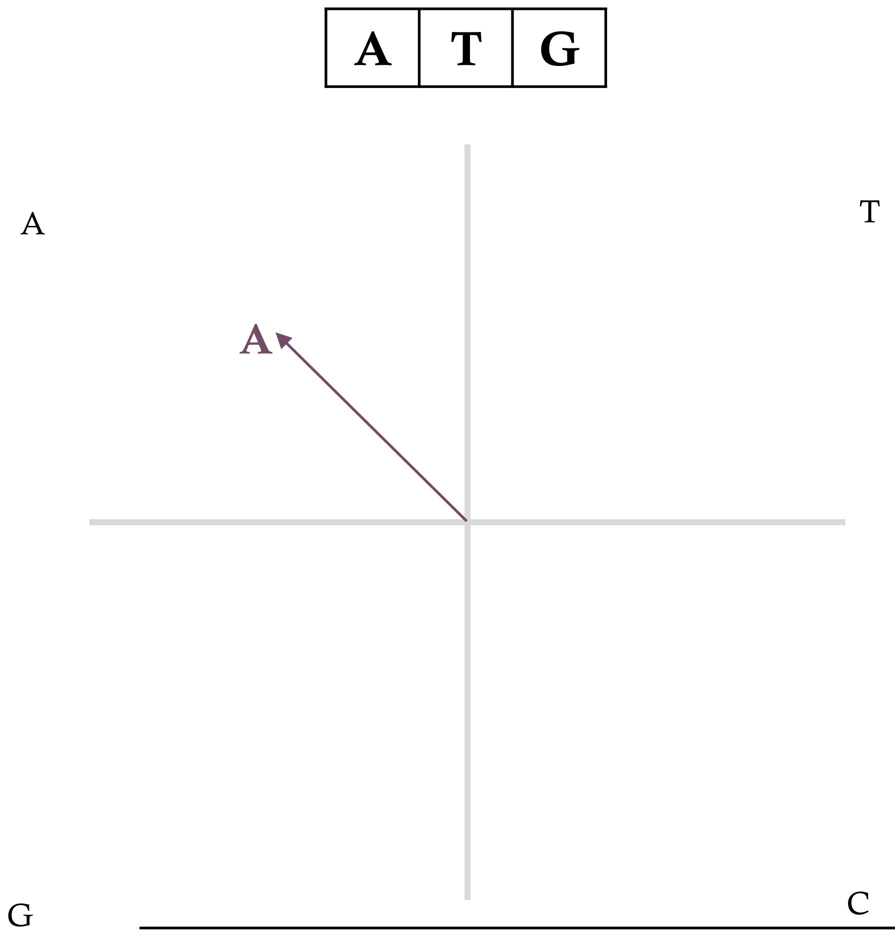

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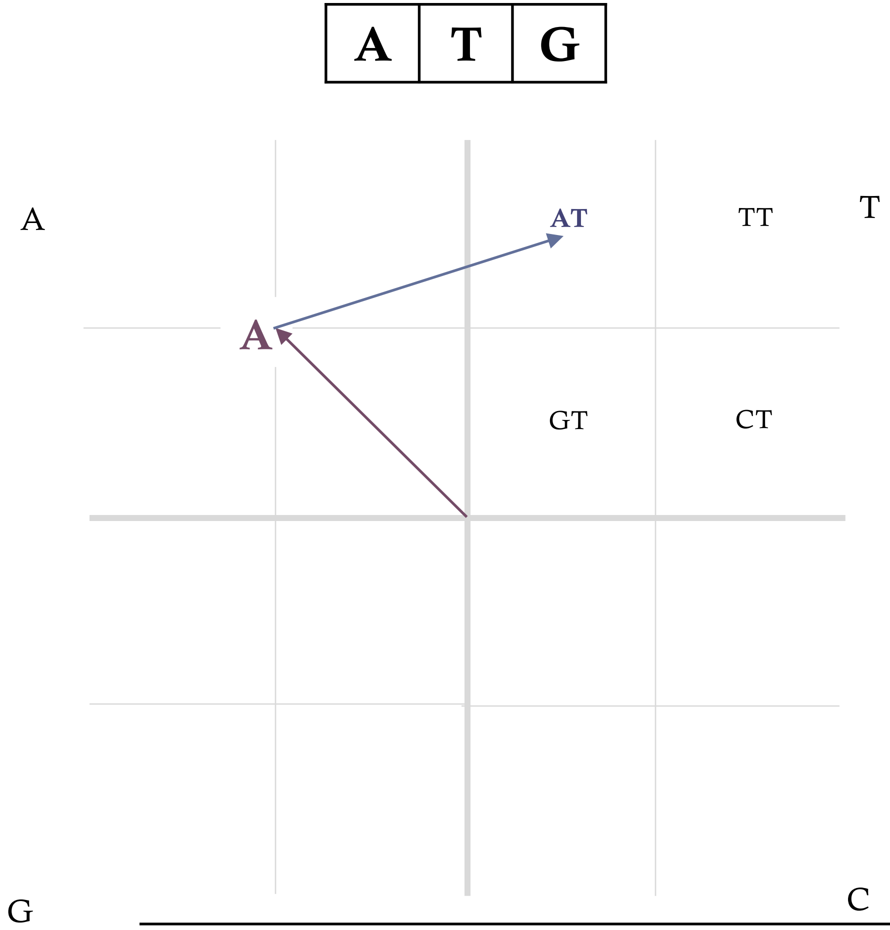

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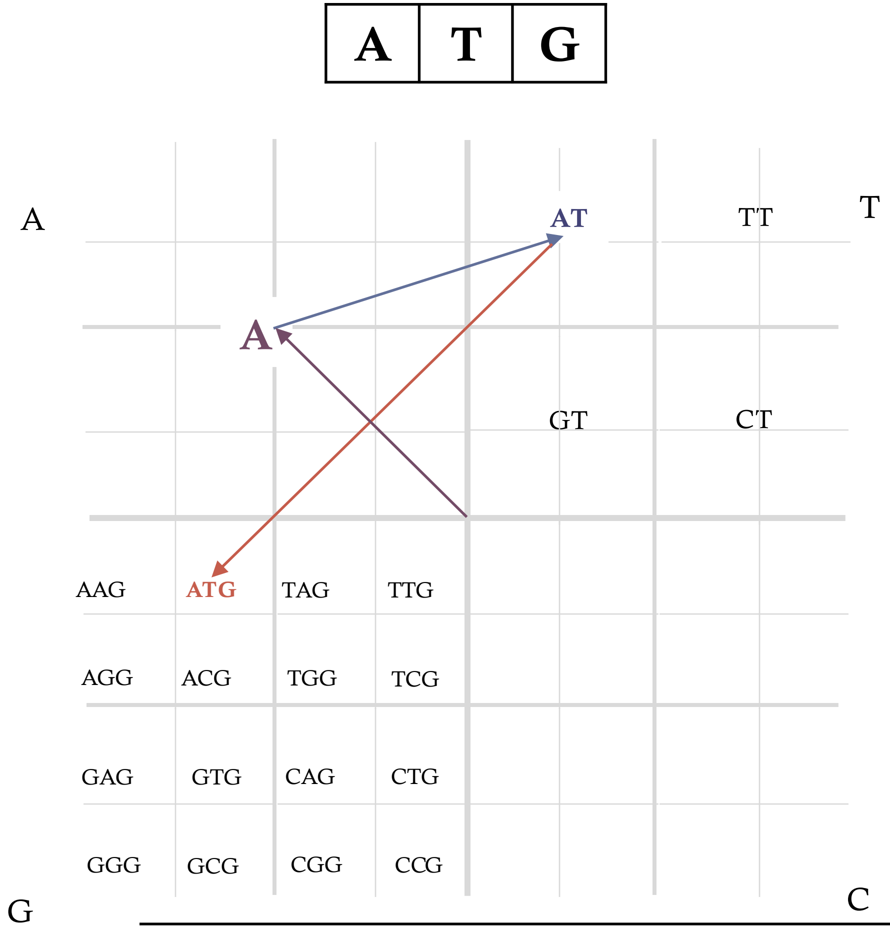

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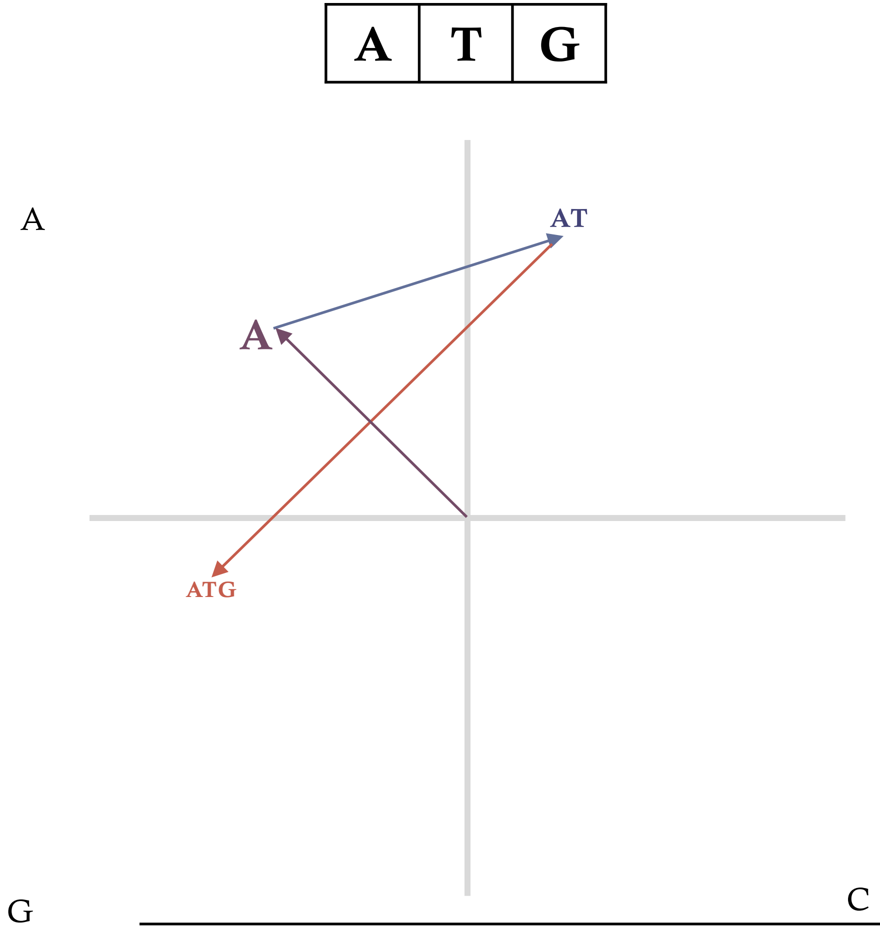

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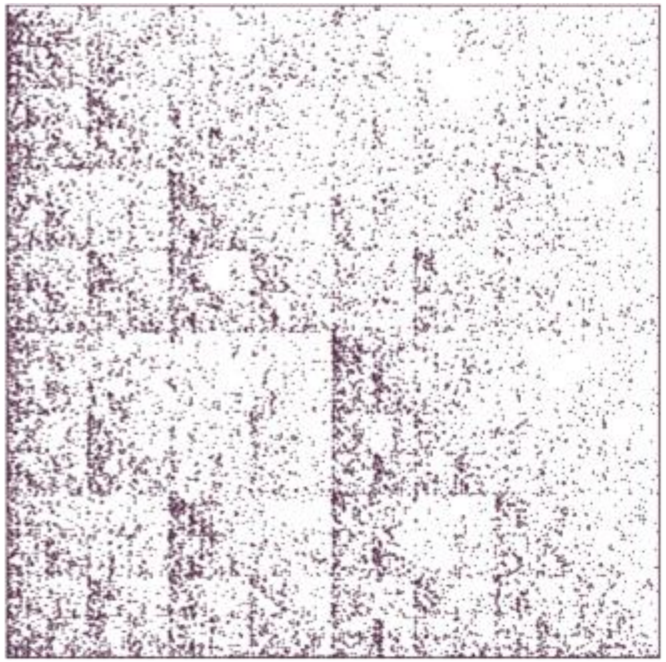

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GRAPH NETWORKS

A	**T**	**G**	**C**	**G**	**C**	**G**	**C**	**G**	**C**

What

Models sequence as a graph/network (e.g., k-mers as nodes, transitions as edges).

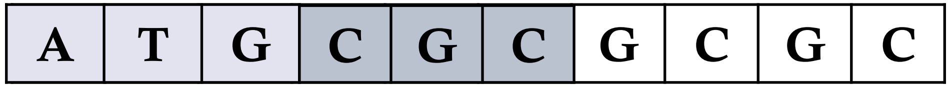
When

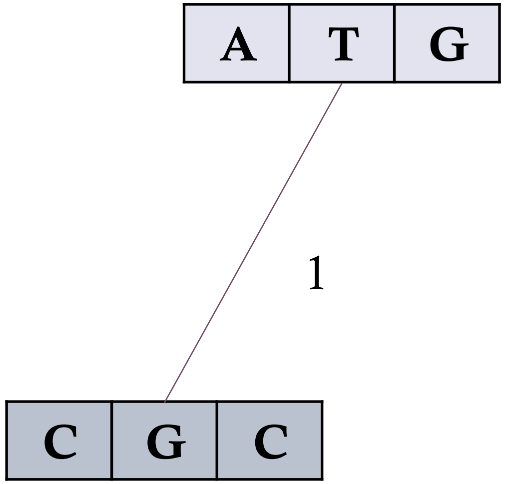
Topological analysis, pattern detection.

Drawbacks

Computationally expensive, less intuitive.

GRAPH NETWORKS





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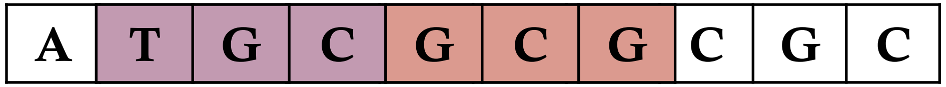

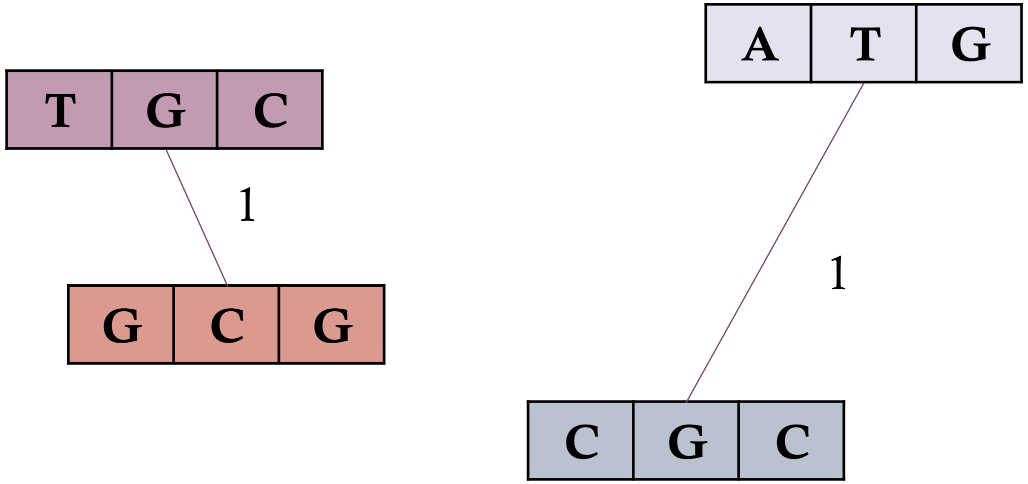

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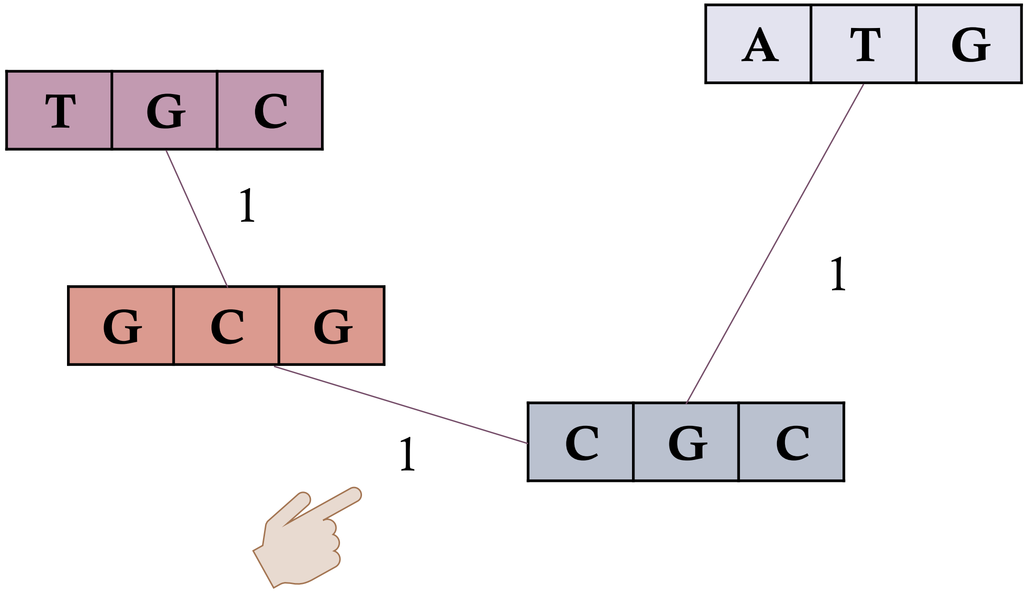

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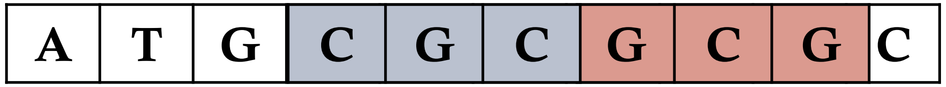

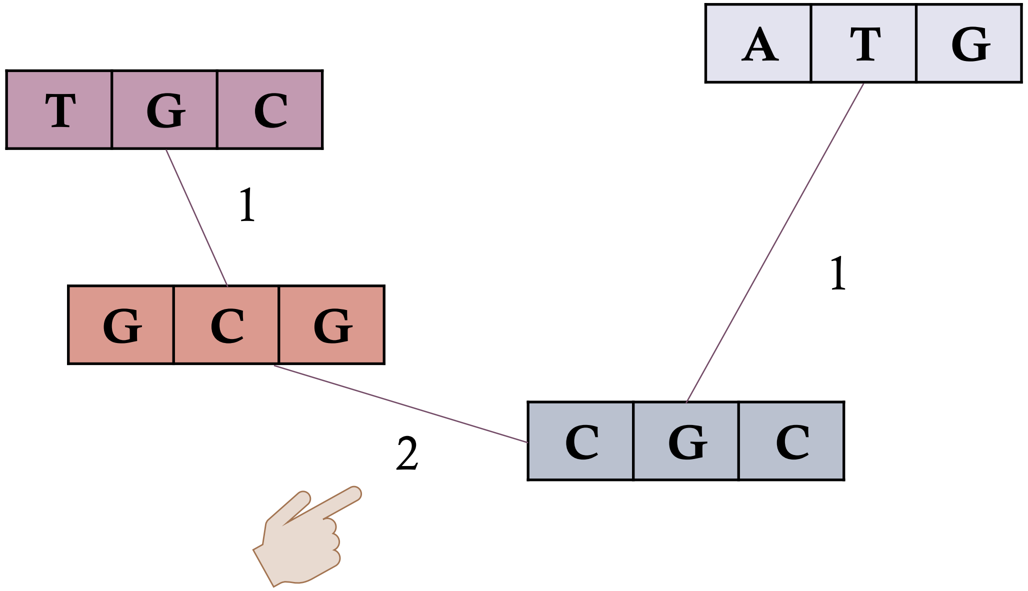

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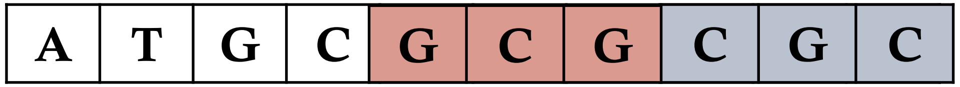

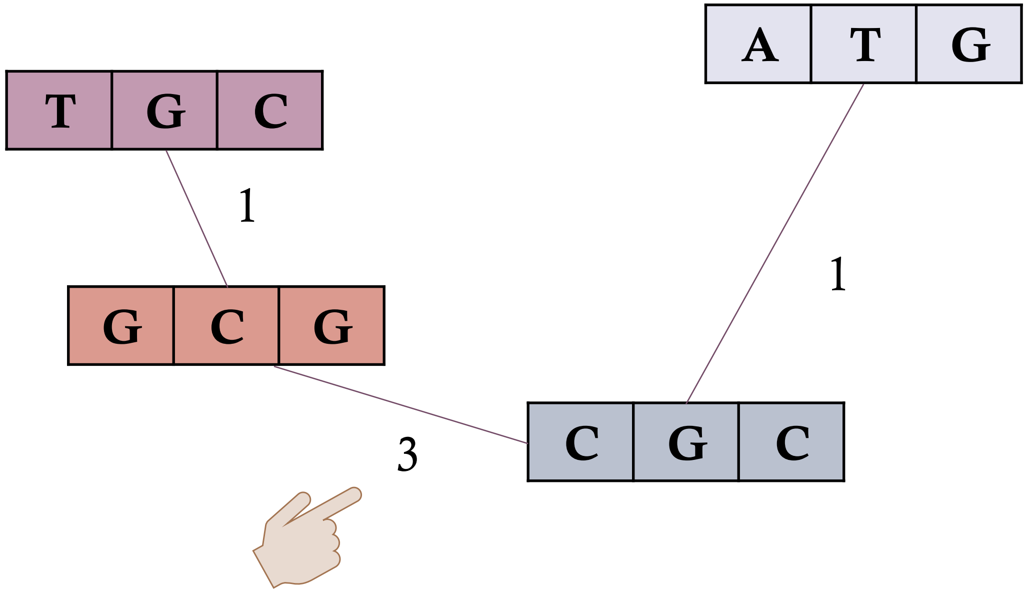

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FEATURE ENGINEERING & MACHINE LEARNING

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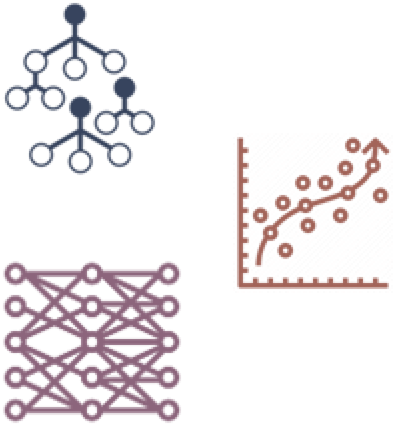
SEQUENCES TO ML-READY INPUT

ATG..
CGC..
GCA..
CAG..
GGA..

1	2	..	..	..	0.5
0	..	..	..	..	1.5
..	..	..	..	..	..
..	..	..	..	..	..
3	..	..	..	..	-0.5



0
1
1
0
1

Convert sequences to numeric representations

Modelling

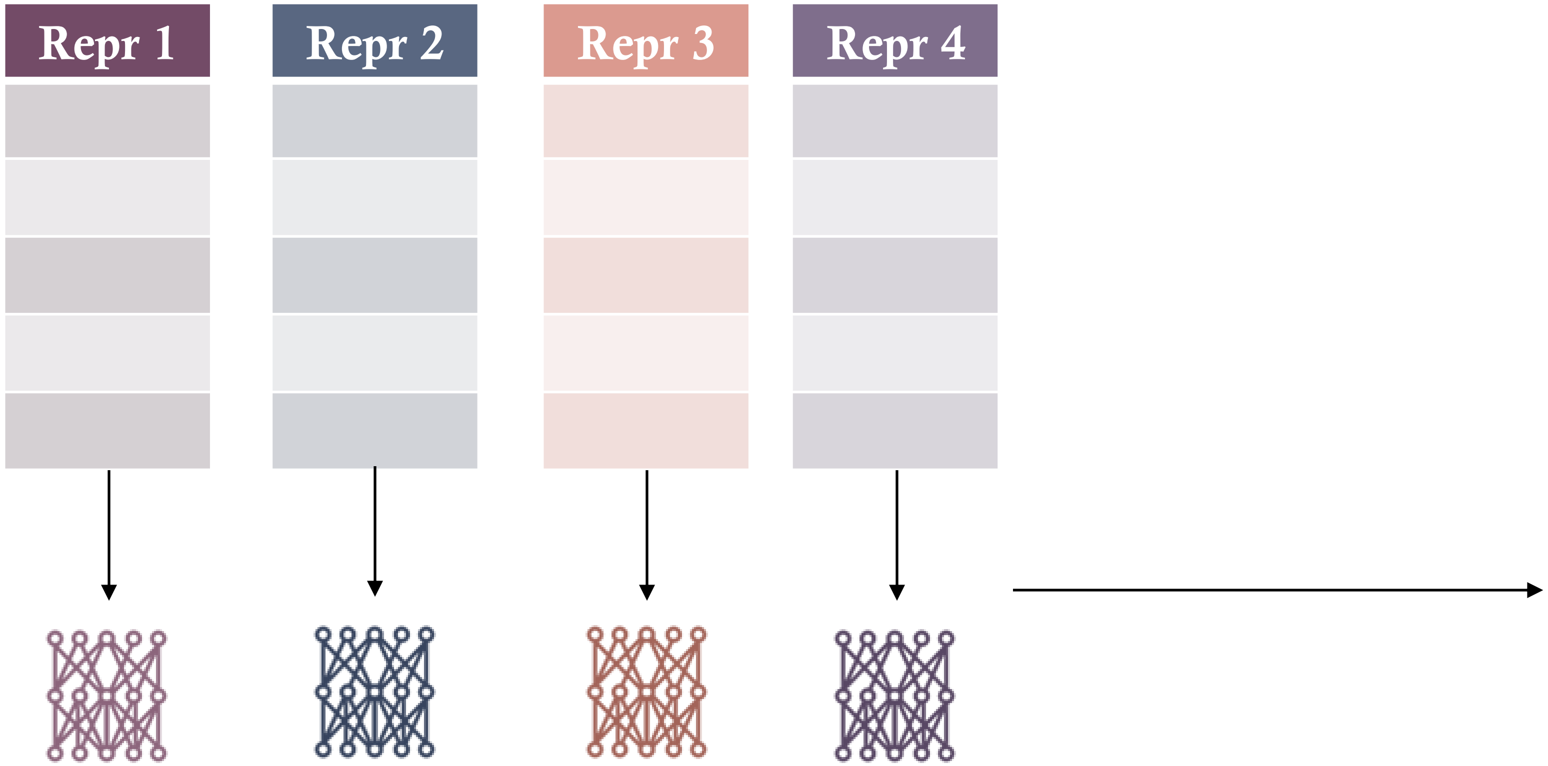
Predictions

FEATURE ENGINEERING

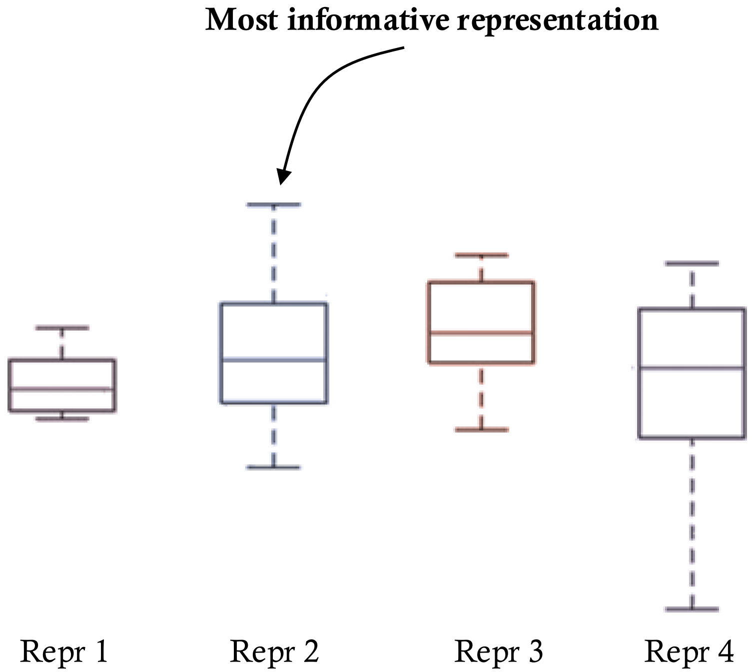
Feature engineering is the process of selecting, creating, or transforming data features to improve the performance of machine learning models.

Repr 1	Repr 2	Repr 3	Repr 4

FEATURE ENGINEERING

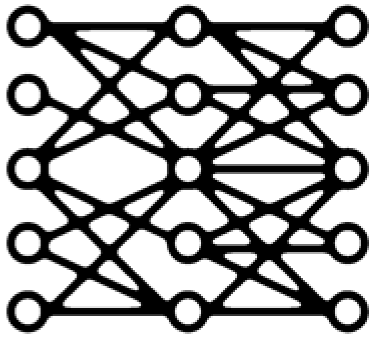


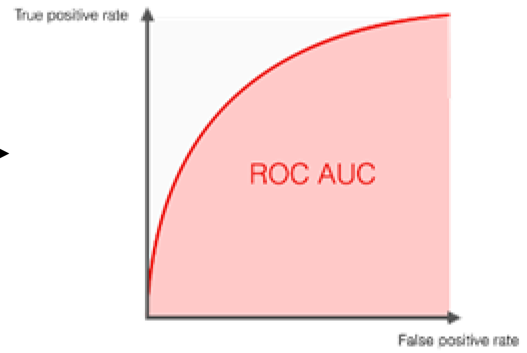
Test each representation individually



FEATURE ENGINEERING

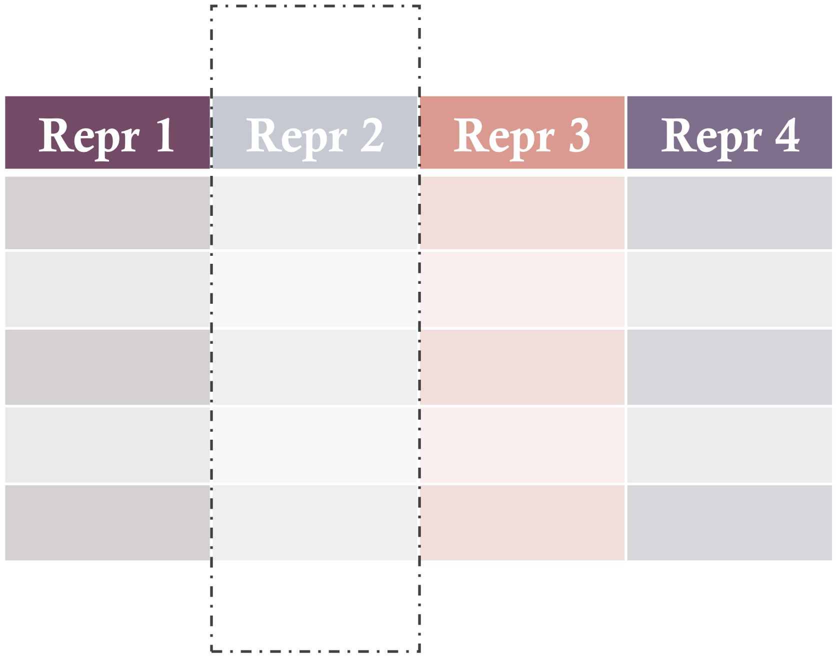
Repr 1	Repr 2	Repr 3	Repr 4



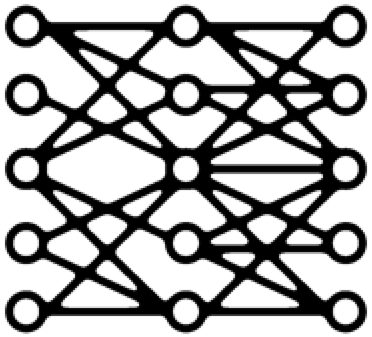


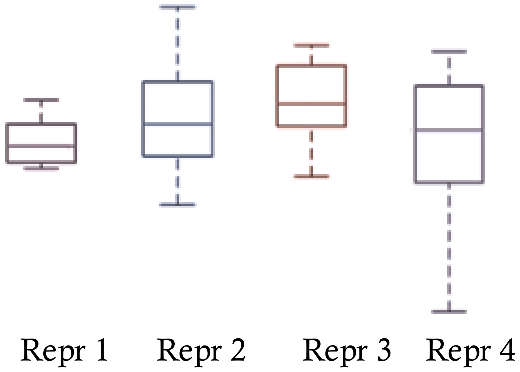
Combine Representations

FEATURE ENGINEERING



Drop each representation from combined data





Effect of Omitting

→ Important features show a drop in performance

SUMMARY

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TYPES OF REPRESENTATIONS

	Name	What	Why/When
Basic	**One-Hot**	Binary representation	Simple and interpretable
	K-mer	Count of K size substrings	Sequence composition analysis
	ORF	Open Reading Frame features	Codon related tasks
Physico-Chemical	**GC Content**	Percentage G/C	Binding/stability
	Nucleotide	Di/Tri nucleotide features	Local sequence features
	Global Descriptor	Global composition	Sequence composition analysis
Numeric Mapping	**Real**	Real number assignment	Simple numeric representation
	Integers	Integer assignment	Simple numeric representation
	Z-curves	3D representation	Reversible, positional context and chemistry
Complex	**Chaos Game**	Fractal representations	Pattern recognition, reversible
	Graphs	Topological representations	Sequence composition and pattern recognition

Complexity

FEATURE ENGINEERING

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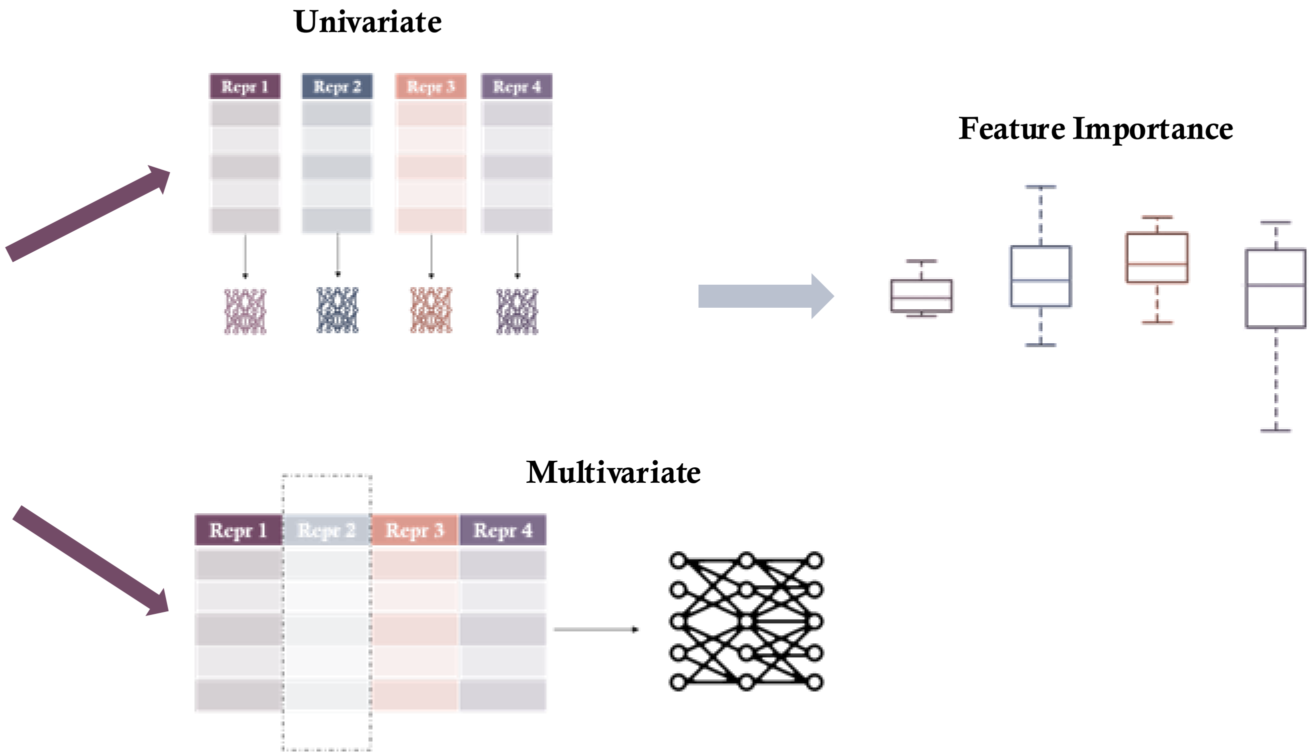

THANKS!
QUESTIONS?

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